

Current Software Defect Prediction: A Systematic Review

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Abstract— detecting defect in a software product prior to testing reduces the cost of testing and improve the quality of the software product. Various methods for enhancing the accuracy of defect prediction model have been published. The goal of this review is to identify and analyze the dataset, models, framework and the performance of software defect prediction model. The IEEEExplore, Science Direct, Scopus, and Google Scholar databases were used to search and download the relevant papers. Sixty-eight (68) papers published between 2017 to 2021 were selected based on exclusion and inclusion criteria. Analysis of the selected studies revealed that 100% of the selected articles used the publicly available dataset from NASA, PROMISE and others. The most frequently used models in software defect prediction studies were identified. The analysis also revealed IEEE Transactions on Software Engineering Journal is the most significant journal with respect to software defect prediction studies using Scimago Journal Ranking as criteria. The review also identified studies on enhancing the predictive performance of defect prediction models. Software defect prediction is still active and sound. Thus, needs more research especially on enhancement methods.

Keywords—Software Defect Prediction, Naïve Bayes, Support Vector Machine, Software

I. INTRODUCTION

Identification of the defect in software product in the early phase of the software development life cycle is very important. Using Software defect prediction (SDP) for identification and removal of software defect contributes greatly in producing cost-effective and qualitative software products. This is due to the fact that, the software defect prediction process is less expensive than software testing and reviews. As reported in some studies, software defect models outperform currently software reviews methods utilized by industries [1]. Furthermore, SDP help software testing teams to focus on the defect prone component. Thus, minimizing the effort of testing and enhancing the quality of software [2], [3]. For these reasons, the SDP has become an important and attractive research topic in software engineering domain. Most of the existing review on SDP studies have not capture the comprehensive image of current approaches such as ensemble techniques and frameworks proposed in studies on software defect prediction. Therefore, in this review work, the datasets, models used, ensemble and frameworks

employed in various studies on software defect prediction models between 2017 and 2021 were identified and analyzed. This systematic review is organized as follows; the methodology is explained in section II, the result of the review is presented in section III and IV. Finally, the conclusion in section V.

A. Research Questions

The research questions addressed by this work are;

- RQ1. Which of the journals is the most significant in the SDP area?
- RQ2. Who are the most influential and active researchers in SDP?
- RQ3. What are research topics in SDP area?
- RQ4. What are the datasets mostly used in SDP?
- RQ5. What are the models used for SDP?
- RQ6. What are the most commonly used models for SDP?
- RQ7. What are the best performing models for SDP?
- RQ8. What are the improvement methods proposed for SDP?
- RQ9. What are the frameworks proposed for SDP?
- RQ10. What are the feature selection methods proposed in SDP?
- RQ11. What are the ensemble methods proposed in SDP?

II. METHOD

The work followed the guideline proposed by [4]. as shown in Fig.1. The planning stage includes; identifying reasons for the review, the formation of research questions, designing and following the suitable method for searching relevant materials, extracting and synthesizing of information for performing the review and as well as evaluation. The next step is performing the review where the searching method, extraction and data synthesis are implemented.



Fig. 1. Systematics Literature review process

A. Search process

The search process begins by defining search strings, selecting databases and finally stating the inclusion and exclusion criteria. Alternative spellings, synonyms and Boolean ANDs and ORs were considered for constructing search strings. Digital libraries such as IEEEExplore, Science Direct, Scopus and Google Scholar were searched.

B. Inclusion and Exclusion criteria

Peer-reviewed articles SDP published between 2017 and 2021 were included:

Inclusion Criteria;

Empirical studies on SDP, study comparing the performance of models in SDP, Studies that used feature selection technique in SDP, Studies that used ensemble methods for SDP were all considered

Exclusion Criteria

Studies without experiment or empirical analysis and studies in any language other than English

C. Assessment of the Quality

Each article was evaluated based on stated assessment of quality questions as shown in Table II. The articles were score as follows; for each question, the score of 1 was given to Yes, 0.5 to Partial and 0 to No.

D. Collection of Data

The extracted data from each article were:

- Authors
- Titles
- Publishers
- Independent variable
- Datasets
- Evaluation metrics
- Models

E. Data analysis

MS excel was used to tabulate the data extracted from each article to show;

- Number of articles published per year
- Number of articles published per author
- Number of articles per model
- Quality score for each article

TABLE I. QUALITY ASSESSMENT

S/N	Question	Yes	Partially	NO
1	Are the study's objectives clearly stated?			
2	Is the study scope clearly stated?			
3	Are there relevant literature in the study?			
4	Are the dependent variable(metrics) clearly stated			
5	Are the independent variable clearly stated?			
6	Are the dataset used clearly presented?			
7	Has the study been cited?			
8	Are the performance evaluation techniques properly written?			

III. RESULTS

A. Search result

A total of 2609 articles were identified initially, after applying exclusion criteria, some based on the title, some based on abstract and finally based on content and removal of duplicate 68 articles were considered.

B. Quality Evaluation of Selected Articles

As shown in Table I, articles scored 5 or above were selected otherwise, rejected. A nomenclature was assigned to each study i.e., PS (Primary Study). After quality assessment, 68 studies were considered and named as primary studies.

IV. DISCUSSION

Response to research questions is discus in this section.

A. Significant Journal Publications

Sixty-eight (68) studies on software defect prediction are considered in this review. Fig. 2 shows the distribution of selected studies per year. As can be seen from the Fig. 2, most of the studies were published in 2020. Fig. 2 also shows that, the relevancy of research on SDP is sound and active till today. To address the research question RQ1: the most significant journal in selected primary studies, Scimago ranking metrics such as SJR value and Quartiles categories from Q1 to Q4 were used to find the most significant as shown in Table II. IEEE Transactions on Software Engineering emerge the most significant journal.

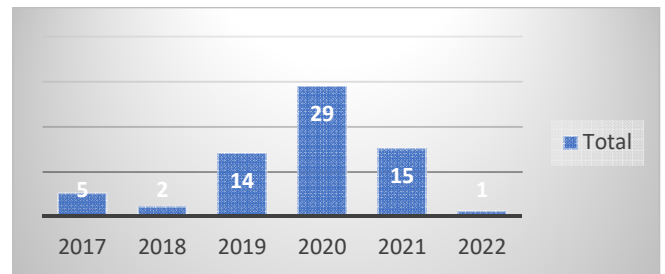


Fig. 2. Distribution of study per year

B. Most Active and Influential Researchers

The researchers in the area of SDP who contributed a lot in the selected papers were identified based on the number of studies they appeared as first Autor. Balogun A.O. contributed as the first Autor in four studies followed by Yu Q and Zhang N who contributed in three (3) studies and as the first Autor in two (2) studies each. Shown in Fig. 3.

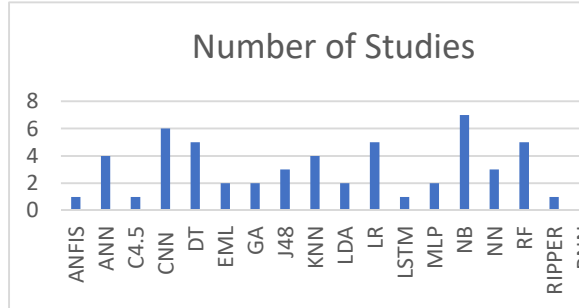


Fig. 4. Distribution of the number of studies per model

C. Research Topics in the Area of Software Defect Prediction

Analysis of selected studies showed that recent research in software defect prediction concentrate on four topics:

- Just-in-time SDP: The major aim of this topic is the identification of defect inducing changes.
- Defect analysis: This topic involves classifying defect into different categories and identifying the possible causes of the defect so as to prevent them from reoccurring.
- Cross Version Defect Prediction (CVDP): in this topic, defect labels of the current version of software is predicted using the classification algorithm trained on the historical dataset of the previous version of the same software.
- Cross-project defect prediction (CPDP): major aim of this topic is to predict the defect labels of one project is predicted using classification model trained on the historical data of another project.

D. Datasets Used for SDP Studies

A dataset is a group of data. Dataset is used for training and evaluating the SDP model. The major component in SDP study is dataset. Public datasets are obtained from NASA, PROMISE, AEEEM, Relink, SOFTLAB, etc. Most of the selected studies used the combination of multiple datasets for analysis.

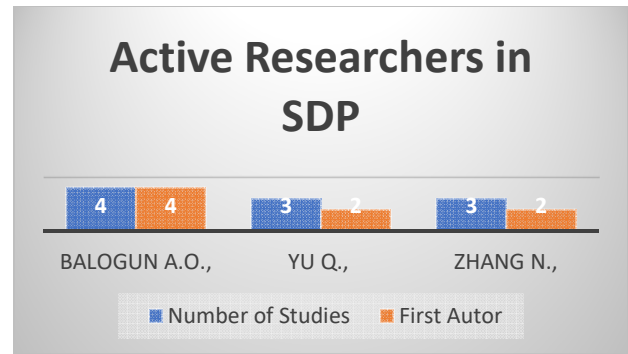


Fig. 3. Distribution of Influential Researchers and Number of Studies

TABLE II. RANKING OF SELECTED JOURNALS BASED ON SCIMAGO JOURNAL RANKING (SJR)

Journal Publication	SJR	Quartile
IEEE Transactions on Soft. Eng.	857	Software (Q1)
Journal of King Saud University – Comp. and Infor. Sci.	617	Comp. Sci. (miscellaneous) (Q1)
Information and Software Technology	606	Software (Q2)
International Journal of Computational Intelligence Systems	385	Computer Science (miscellaneous) (Q2)
Symmetry	385	Computer Science (miscellaneous) (Q2)
IET Software	305	Computer Graphics and Computer-Aided Design (Q3)
Journal of Systems Engineering and Electronics	301	Computer Applications (Q3)
International Journal of Electrical and Computer Engineering	277	Computer Science (miscellaneous) (Q2)

E. Methods(Models) Used in SDP Studies

As shown in Fig. 4, the most useful models are NB and SVM.

F. Most Frequently Used Methods(Models) in SDP Studies

From the used methods in software defect prediction in Section E, six most frequently used models in SDP were identified based on the number of studies they were employed in. Naïve Bayes (NB) and Support Vector Machine (SVM) emerged as the most useful methods followed by Convolutional Neural Network (CNN).

G. Best Performing Models for SDP

Two out of selected studies reported RF and SVM as better performing models such studies are; [5] and [6].

H. Proposed Improvements Method for SDP models

In order to improve the SDP models' predictive performance, researchers focus on improving the quality of

training and testing datasets (using feature selection techniques), optimizing classifiers (ensemble techniques) or the combination of both.

1) Feature Selection

Different researchers concentrated their studies toward improving SDP models through feature selection techniques such as studies performed by [7], [8], [9], [10], [11], [12] and [13]

2) Ensemble Techniques

On the other hand, some studies concentrate on using the ensemble method to improve software defect prediction. Ensemble method is a procedure in which multiple classifiers are combined together in order to achieve better performance. In addition to improving classifiers' accuracy, ensemble method has been used for mitigating the issue of class imbalance in software defect prediction. Examples of such studies are [14], [15], [16], [17] and [18].

I. Proposed Frameworks for Software Defect Prediction

Various studies proposed different framework to improve SDP model. Below are the frameworks proposed for software defect prediction.

[19] proposed a framework which consists three layers; the first layer is the pre-processing layer where missing values and outliers in the dataset are handled, and dataset are split in to training and testing data. The classification through four different back-propagation techniques such as BR, LM, SCG and BFGS-QN was done in the second layer. The third layer includes fuzzy technique where the best training algorithms selected based on the performance and stored in the cloud so that the development team can access the model anywhere. Thus, reducing the cost of testing. The framework was evaluated on NAA dataset using various metrics like Recall, Precision, Accuracy, F-measure, AUC, R2 and MSE. It was observed that the BR training function outperformed the other training functions.

[20] proposed an integrated machine learning framework for Infrastructure as a Code (IaC) defect prediction which allows crawling of the repository, extraction of metrics, building model and evaluation. Five machine learning algorithms and 104 projects were used to evaluate the performance of the framework in identifying defective IaC scripts. The result of the experiment shows that Random Forest outperformed the other classification algorithm especially for AUC and MCC evaluation metrics.

[21] proposed a framework to solve the issue of class imbalance SDP models. In this approach, a balance between the defect and non-defect instances in the imbalance dataset are created using the distribution properties of the dataset. Experiment result shows that, the proposed framework outperformed SMOTE and K-Means SOMTE.

[22] to improve the performance of SDP, this study proposed a framework in which the variant selection activity was used to identify the best optimized version of

classification techniques was combined with feature selection technique. Based on evaluation, the framework outperformed some commonly used classification techniques.

[23] proposed a framework which improve the performance of Extreme Machine Learning (EML). The framework comprises three components; component 1: KPCA leverages a nonlinear mapping function to extract optimal representative features. Component 2: leverages adaptive genetic algorithm to enhance the prediction performance of EML. Component 3: the multiple EML optimized by the adaptive genetic algorithm re integrated by Adaboost algorithm. The framework was evaluated on eleven open-source datasets and compared with some machine learning techniques, including EML and three other variants. The result shows that, the proposed framework outperformed the base line models.

[24] proposed a framework which consist of three stages; stage1: the influence of the minority class in the dataset is minimized using Sampling with Minority Method (SWIM) based on *Mahalanobis* distance technique. Stage2: the most relevant representative feature between the training and the testing dataset was selected. Stage3: performs the transfer learning from the training to the testing dataset in the Grossmann manifold. The experimental result shows that, the proposed framework is superior in terms of AUC.

[25] proposed a framework which identify defect using features extracted from the commit automatically when messages and code changes. Experiment result shows that, the approach achieves 10.36 to 11.02% improvement over the state-of-the-art methods.

J. Limitations of this study

Only open access papers were downloaded out of which 68 studies were selected for this review and as such, there might have been an exclusion of some relevant software defect prediction studies from journals or conference proceedings.

V. CONCLUSION

The aim of systematic review was to identify and analyze current models used, datasets, best performing models, topics, improvement methods proposed and frameworks in SDP. Analysis performed on the selected primary studies revealed that, 100% of the selected papers utilized public datasets for analysis. RF and SVM performed better than most of the classification algorithms in SDP. The feature selection techniques play a vital role in improving the quality of the dataset used for SDP likewise, ensemble techniques have been used for enhancing the predictive performance of SDP models especially weak classifiers.

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TABLE III. LIST OF SELECTED PRIMARY STUDIES

Key	Authors		
PS1	Daoud et al (2022)	PS35	Zheng et al (2020)
PS2	Ali et al (2021)	PS36	Ali et al (2020)
PS3	Balogun et al (2021)	PS37	Miholca et al (2020).
PS4	Mustaqeem & Saqib (2021)	PS38	Wattanakriengkrai et al (2020)
PS5	Pan et al (2021)	PS39	Zhang et al (2020)
PS6	Xu et al (2021).	PS40	Gong et al (2020)
PS7	Hayakawa et al (2021)	PS41	Zhang et al (2020)
PS8	Balogun et al (2021)	PS42	Iqbal & Aftab (2020)
PS9	Mumtaz et al (2021).	PS43	Khan (2020).
PS10	Yu et al (2021)	PS44	Jiang et al (2020)
PS11	Amara et al (2021)	PS45	Albahli (2020)
PS12	Wang et al (2021)	PS46	Balogun et al (2019)
PS13	Sikic et al (2021)	PS47	Kakkar et al (2019).
PS14	Lin & Lu (2021).	PS48	Yu et al (2019)
PS15	Dallapalma et al (2021)	PS49	Cabral et al (2019)
PS16	Pandey et al (2021)	PS50	Paterson et al (2019)
PS17	Bennin et al (2020)	PS51	He et al (2019)
PS18	Saifan & Abuwardih (2020).	PS52	Oztürk (2019)
PS19	Bashir & Yahaya (2020)	PS53	Jian et al (2019)
PS20	Morasca & Lavazza (2020)	PS54	Hammad et al (2019)
PS21	Yucalar et al (2020)	PS55	Zheng et al (2019)
PS22	Zhang et al (2020)	PS56	Liang et al (2019)
PS23	Aquil & Ishak (2020).	PS57	Yang et al (2019)
PS24	Balogun et al (2020).	PS58	Ji et al (2019)
PS25	Abaei et al (2020)	PS59	Shao et al (2019)
PS26	Zhu et al (2020)	PS60	Mohapatra & Ray (2018)
PS27	Yu et al (2020).	PS61	Sunil et al (2018)
PS28	Murakami et al (2020)	PS62	Felix & Lee (2017)
PS29	Rhmann et al (2020)	PS63	Ma et al (2017)
PS30	Ren & Liu (2020)	PS64	Ibarguren et al (2017)
PS31	Bejjanki et al (2020).	PS65	Yu et al (2017)
PS32	Alsawalqah et al (2020)	PS66	Wan et al (2017)
PS33	Alqadi & Maletic (2020)	PS67	Iqbal et al (2019)
PS34	Naseem et al (2020)	PS68	Khan et al. (2021)

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